```
------------------------------------------------------------------------------------
> ------------------------------------------------------------------
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-1.log
  log type:  text
 opened on:  14 May 2026, 15:03:41

. 
. clear all

. set seed 1000

. 
. * Task 1:
. *--------------------------------------------------------------------------------
. 
. 
. import excel "USD to BDT.xlsx", firstrow clear
(2 vars, 4,480 obs)

. 
. rename Close Ex_Rate

. label variable Ex_Rate "Exchange Rate"

. 
. replace Date = dofc(Date)
(4,480 real changes made)

. format Date %td

. tsset Date

Time variable: Date, 03jan2011 to 31dec2025, but with gaps
        Delta: 1 day

. 
. 
. * Task 2:
. *--------------------------------------------------------------------------------
. set scheme cleanplots

. 
. tsline Ex_Rate, ///
>        lcolor(navy) lwidth(medthick) ///
>            title("{bf:Daily Exchange Rate Trend (2011-2025)}", size(large)) ///
>            xtitle("Year", size(medlarge)) ///
>            ytitle("Rate (BDT)", size(medlarge)) ///
>            ylabel(, labsize(medium)) ///
>            tlabel(01jan2011 01jan2016 01jan2021 01jan2026, format(%tdcY) labsize(med
> ium)) ///
>            graphregion(color(white)) xsize(8) ysize(4) ///
>            name(tsplot, replace) note("Source: Google Finance")

. 
. graph export Hw_1_tsplot.emf, name(tsplot) replace
file F:\ECON 408\Practice do file\Hw_1_tsplot.emf saved as Enhanced Metafile format

. 
. 
```

```
. * Task 3:
. *--------------------------------------------------------------------------------
.
. bcal create bus_cal, from(Date) gen(time) replace

  Business calendar bus_cal (format %tbbus_cal):

    Purpose:

      Range: 03jan2011  31dec2025
                 18630      24106   in %td units
                     0       4479   in %tbbus_cal units

     Center: 03jan2011
                 18630              in %td units
                     0              in %tbbus_cal units

    Omitted:       997              days
                    66.5            approx. days/year

   Included:     4,480              days
                   298.8            approx. days/year

  Notes:

    Business calendar file bus_cal.stbcal saved.

    Variable time created; it contains business dates in %tbbus_cal format.

. format time %tbbus_cal

. tsset time

Time variable: time, 03jan2011 to 31dec2025
        Delta: 1 day

. gen Diff_Ex = D.Ex_Rate
(1 missing value generated)

.
.
. * Task 4:
. *--------------------------------------------------------------------------------
.
. * Step 1: Stationarity Check
.
. tsset Date

Time variable: Date, 03jan2011 to 31dec2025, but with gaps
        Delta: 1 day

.
. set scheme cleanplots

.
. tsline Diff_Ex, ///
>        lcolor(navy) lwidth(medthick) ///
>            title("{bf:Daily Diff_Ex Plot (2011-2025)}", size(large)) ///
>            xtitle("Year", size(medlarge)) ///
>            ytitle("Rate (BDT)", size(medlarge)) ///
>            ylabel(, labsize(medium)) ///
>            tlabel(01jan2011 01jan2016 01jan2021 01jan2026, format(%tdcY) labsize(med
> ium)) ///
>            graphregion(color(white)) xsize(8) ysize(4) ///
>            name(Diff_Ex_plot, replace)
```

```
.
. graph export Diff_Ex_plot.emf, name(Diff_Ex_plot) replace
file F:\ECON 408\Practice do file\Diff_Ex_plot.emf saved as Enhanced Metafile format

.
. dfuller Diff_Ex

Dickey-Fuller test for unit root        Number of obs  = 3,833
Variable: Diff_Ex                       Number of lags =     0

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test      -------- critical value ---------
              statistic           1%           5%          10%
--------------------------------------------------------------
 Z(t)           -84.872       -3.430       -2.860       -2.570
--------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000.

. tsset time

Time variable: time, 03jan2011 to 31dec2025
        Delta: 1 day

.
. * Step 2: Empirical ACF and PACF check
. set scheme s2color

.
. ac Diff_Ex, name(acf_plot,replace) lags(20)

. pac Diff_Ex, name(pacf_plot,replace) lags(20)

. graph combine acf_plot pacf_plot, title("ACF and PACF of Diff_Ex") xsize(8) ysize(4)
>  name(acf_pacf_plot, replace)

.
. graph export hw_1_acf_pacf_plot.emf, name(acf_pacf_plot) replace
file F:\ECON 408\Practice do file\hw_1_acf_pacf_plot.emf saved as Enhanced Metafile fo
> rmat

.
. *ARMA(2,3) , ARMA(3,2) , ARMA(3,3) , ARMA(3,4), ARMA(4,3)
.
.
. qui arima Diff_Ex, ar(1/2) ma(1/3)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .  -3029.567       7   6073.134   6117.984
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

.
```

```
. qui arima Diff_Ex, ar(1/3) ma(1/2)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .    -3028.1       7   6070.201   6115.051
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

.
. qui arima Diff_Ex, ar(1/3) ma(1/3)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .  -3027.874       8   6071.749   6123.006
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

.
. qui arima Diff_Ex, ar(1/3) ma(1/4)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .  -3027.815       9    6073.63   6131.295
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

.
. qui arima Diff_Ex, ar(1/4) ma(1/3)

. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .  -3027.432       9   6072.865   6130.529
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

.
.
. * ARMA(3,2) Model has the lowest AIC and BIC
.
. * Step 3: Estimation
```

```
.
. arima Diff_Ex, ar(1/3) ma(1/2)

(setting optimization to BHHH)
Iteration 0:   log likelihood = -3040.2677
Iteration 1:   log likelihood =  -3031.423
Iteration 2:   log likelihood = -3031.0705
Iteration 3:   log likelihood = -3030.8361
Iteration 4:   log likelihood =  -3030.624
(switching optimization to BFGS)
Iteration 5:   log likelihood = -3030.4353
Iteration 6:   log likelihood = -3028.2888
Iteration 7:   log likelihood = -3028.2146
Iteration 8:   log likelihood =  -3028.186
Iteration 9:   log likelihood = -3028.1531
Iteration 10:  log likelihood = -3028.1209
Iteration 11:  log likelihood = -3028.1159
Iteration 12:  log likelihood = -3028.1059
Iteration 13:  log likelihood =  -3028.104
Iteration 14:  log likelihood = -3028.1029
(switching optimization to BHHH)
Iteration 15:  log likelihood = -3028.1018
Iteration 16:  log likelihood =  -3028.101
Iteration 17:  log likelihood = -3028.1009
Iteration 18:  log likelihood = -3028.1008
Iteration 19:  log likelihood = -3028.1008
(switching optimization to BFGS)
Iteration 20:  log likelihood = -3028.1007
Iteration 21:  log likelihood = -3028.1005
Iteration 22:  log likelihood = -3028.1004
Iteration 23:  log likelihood = -3028.1003
Iteration 24:  log likelihood = -3028.1003
Iteration 25:  log likelihood = -3028.1003
Iteration 26:  log likelihood = -3028.1003
Iteration 27:  log likelihood = -3028.1003
Iteration 28:  log likelihood = -3028.1003
Iteration 29:  log likelihood = -3028.1003

ARIMA regression

Sample: 04jan2011 thru 31dec2025                Number of obs     =       4479
                                                Wald chi2(5)      =   98684.67
Log likelihood = -3028.1                        Prob > chi2       =     0.0000

------------------------------------------------------------------------------
             |                 OPG
     Diff_Ex | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Diff_Ex      |
       _cons |   .0114831   .0039755     2.89   0.004     .0036914    .0192749
-------------+----------------------------------------------------------------
ARMA         |
          ar |
         L1. |   .5417597    .065313     8.29   0.000     .4137485    .6697709
         L2. |     -.2241   .0181753   -12.33   0.000    -.2597229    -.188477
         L3. |  -.1330928   .0111702   -11.92   0.000    -.1549859   -.1111997
             |
          ma |
         L1. |  -1.022842    .066048   -15.49   0.000    -1.152293   -.8933901
         L2. |   .3989004   .0464674     8.58   0.000     .3078259    .4899748
-------------+----------------------------------------------------------------
      /sigma |   .4757236    .001023   465.02   0.000     .4737185    .4777287
------------------------------------------------------------------------------
Note: The test of the variance against zero is one sided, and the two-sided
      confidence interval is truncated at zero.
```

```
. estat ic

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
           . |      4,479          .    -3028.1       7   6070.201   6115.051
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] BIC note.

. estat aroots , name(arootshw1, replace)

   Eigenvalue stability condition
  +----------------------------------------+
  |        Eigenvalue        |   Modulus   |
  |--------------------------+-------------|
  |   .4147067 +  .5391681i  |   .680209   |
  |   .4147067 -  .5391681i  |   .680209   |
  |  -.2876538               |   .287654   |
  +----------------------------------------+
   All the eigenvalues lie inside the unit circle.
   AR parameters satisfy stability condition.

   Eigenvalue stability condition
  +----------------------------------------+
  |        Eigenvalue        |   Modulus   |
  |--------------------------+-------------|
  |   .5114209 +  .3706063i  |   .631586   |
  |   .5114209 -  .3706063i  |   .631586   |
  +----------------------------------------+
   All the eigenvalues lie inside the unit circle.
   MA parameters satisfy invertibility condition.

. graph export arootshw1.emf, name(arootshw1) replace
file F:\ECON 408\Practice do file\arootshw1.emf saved as Enhanced Metafile format

. predict res , residuals
(1 missing value generated)

.
. ac res, name(acf_res,replace) lags(10)

. pac res, name(pacf_res,replace) lags(10)

. graph combine acf_res pacf_res, title("ACF and PACF of Residuals") xsize(8) ysize(4)
>  name(hw1_res_ac_pac_plot,replace)

. graph export hw1_res_ac_pac_plot.emf, name(hw1_res_ac_pac_plot) replace
file F:\ECON 408\Practice do file\hw1_res_ac_pac_plot.emf saved as Enhanced Metafile f
> ormat

.
.
. wntestq res, lags(10)

Portmanteau test for white noise
---------------------------------------
 Portmanteau (Q) statistic =    17.7000
 Prob > chi2(10)           =     0.0602

.
.
. log close
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-1.log
  log type:  text
 closed on:  14 May 2026, 15:04:33
-------------------------------------------------------------------------------------
> ------------------------------------------------------------------
```